OCW Humanities LLM Tutor

Project Idea

This project is about building a Socratic LLM tutor for MIT OpenCourseWare humanities and social science courses. The goal is not to make a chatbot that simply gives students answers. Instead, we want the tutor to help students work through assignments by asking useful questions, giving small hints, and pushing them to explain their own thinking.

This is useful because many OCW students study independently. They can access the readings and assignments, but they do not always have someone to guide them when they get stuck. In humanities classes, that guidance matters because the work is usually about interpretation, evidence, and argumentation rather than one final answer.

The main research question is whether an LLM tutor can reliably follow the Socratic method, even when a student is confused, impatient, off-topic, or directly asks for the answer. A second question is how we can test this kind of tutor before real students use it.

Approach

We are building two connected systems. The first is the tutor itself. The tutor uses a versioned prompt that tells it to stay Socratic, keep responses short, ask focused questions, and avoid writing the student's answer. We test it through multi-turn conversations with different student personas.

The second system is the evaluation pipeline. We generate tutor-student conversations, then grade them using a rubric-based LLM judge. The judge checks whether the tutor stayed Socratic, gave helpful scaffolding, stayed connected to the assignment, and communicated clearly. We also built a dashboard so we can compare transcripts, prompt versions, and scores more easily.

So far, we have built the LangGraph conversation pipeline, created 18 student personas, generated about 900 transcripts, added an LLM judge, and built mini-continuation testing. Mini-continuation testing is especially useful because it lets us retest only the part of a conversation where something went wrong instead of regenerating the whole transcript.

Results

Our biggest improvement so far came from moving from tutor_04 to tutor_05. In the mini-continuation tests, tutor_05 did much better at avoiding direct answer-giving. In some earlier cases, the tutor gave fill-in-the-blank style help, which was too close to writing the answer. The newer version more clearly refuses to do the assignment for the student while still giving useful guidance.

One limitation is that the AI student personas are starting to become less realistic. Some simulated students stay confused about simple terms for too long, which a real student probably would not do. Because of this, the next phase should focus more on human testing.

Next Steps

Run human testing with researchers and TAs using the web tutor interface.

Collect examples of where the tutor still fails or becomes too direct.

Create a hand-graded set of about 20 transcripts and compare those scores with the LLM judge.

Use Pearson and Spearman correlation to check whether the judge agrees with human graders.

Add a few more OCW humanities or social science exercises to test whether the tutor generalizes.

Improve the student personas using what we learn from real human testing.

Prepare for the May 12, 2026 soft launch with a small OCW student group.

Expected Progress

By the end of the project, we expect to have a working Socratic tutor, a reusable testing pipeline, a calibrated judge, and early deployment results from an OCW course. The main contribution is not only the tutor, but also the method for checking whether it is actually tutoring in the way we want. If this works, it could help OCW students get more interactive support while still making sure they do the thinking themselves.